

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

Unless otherwise advised by the manufacturer, the following procedures are recommended:

Always renew the timing chain

If the timing chain is to be re-used, mark the running direction before removal

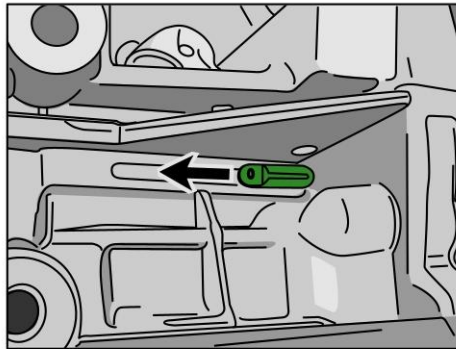
Never turn the engine in the direction opposite to that of normal operation

Note: Avoid bending, torsion or traction

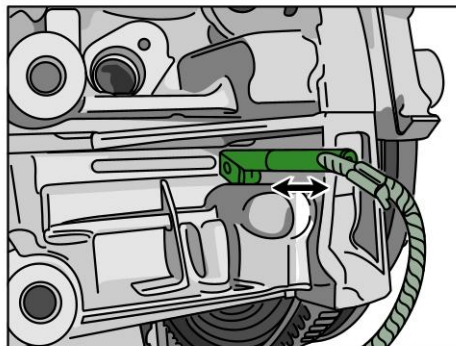
Always use new gaskets

Checking

Remove the access plug



Fit the crankshaft locking tool (11 0 300)

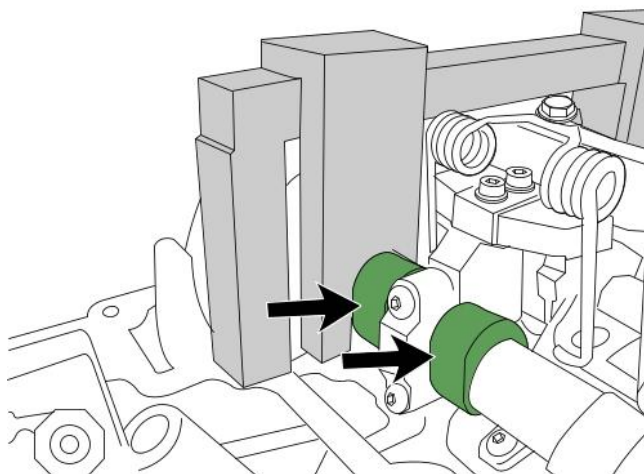


Turn the crankshaft until it touches the TDC locking pin

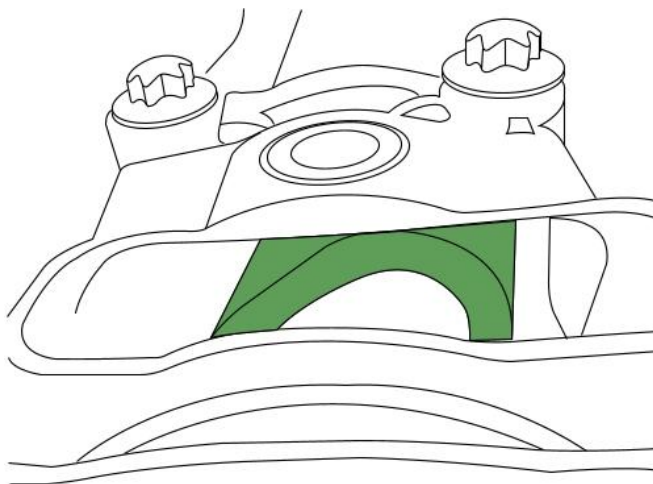
The crankshaft should be at TDC for cylinder 1

Note: Both camshafts must point upwards at an angle

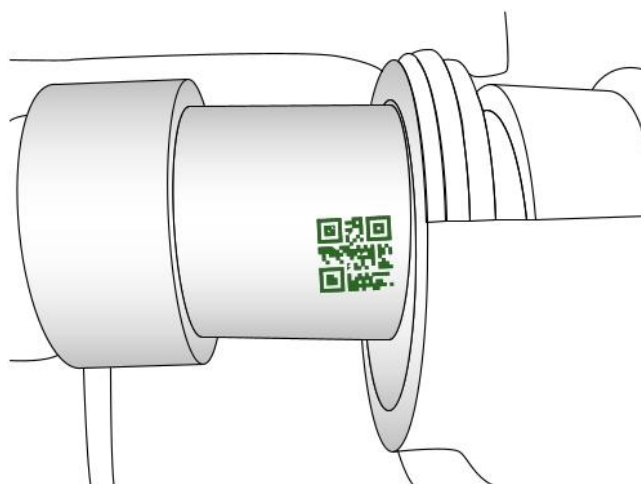
Inlet camshaft cam angle

Repair Manuals

Exhaust camshaft cam angle



Note: Ensure that the marks face upwards

**Removal**

Disconnect the battery

Remove the wiring loom from the cylinder head

Renew the fuel feed lines

Remove the air filter housing and hoses

Repair Manuals

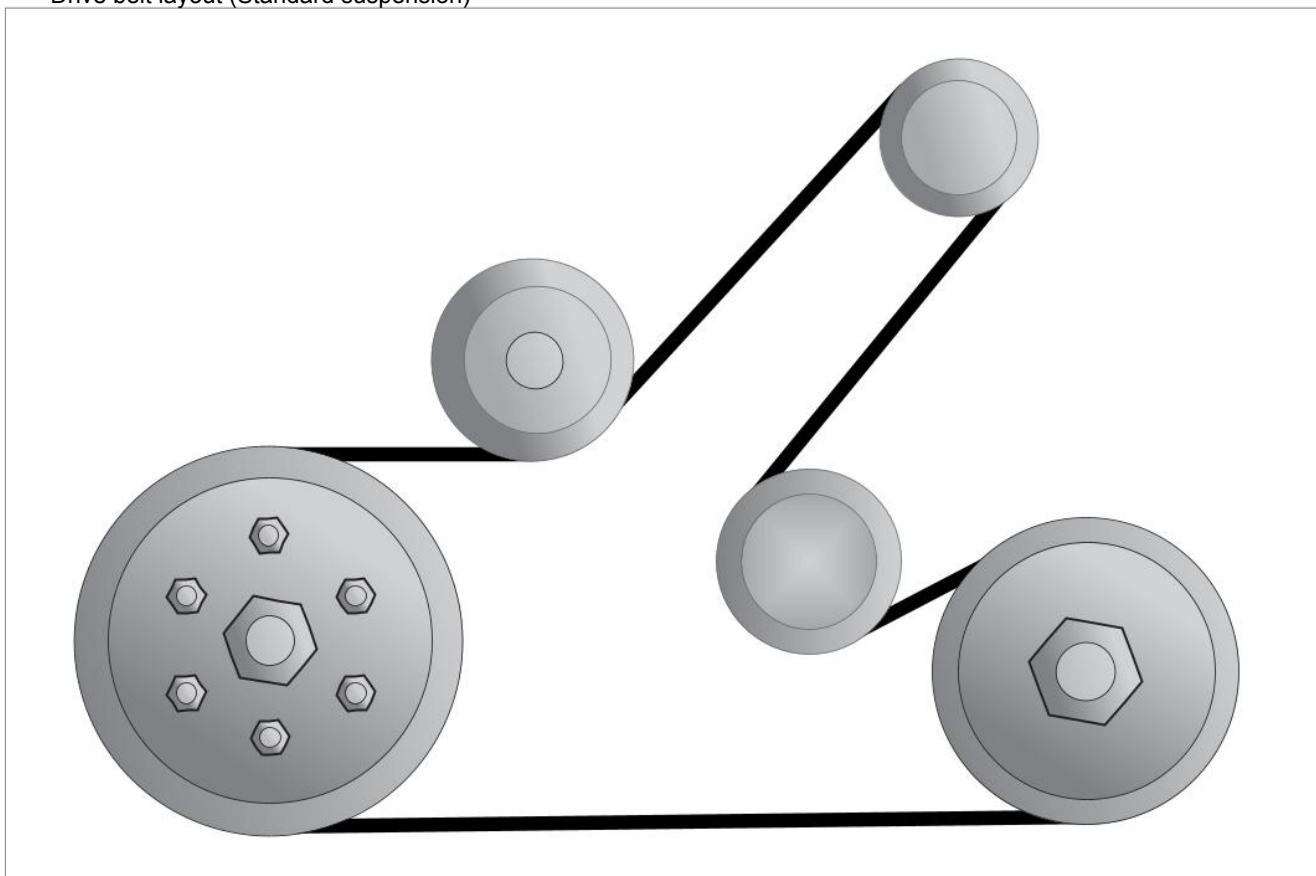
Disconnect all the vacuum hoses from the cylinder head and ancillaries

Remove the cylinder head cover

Remove the spark plugs

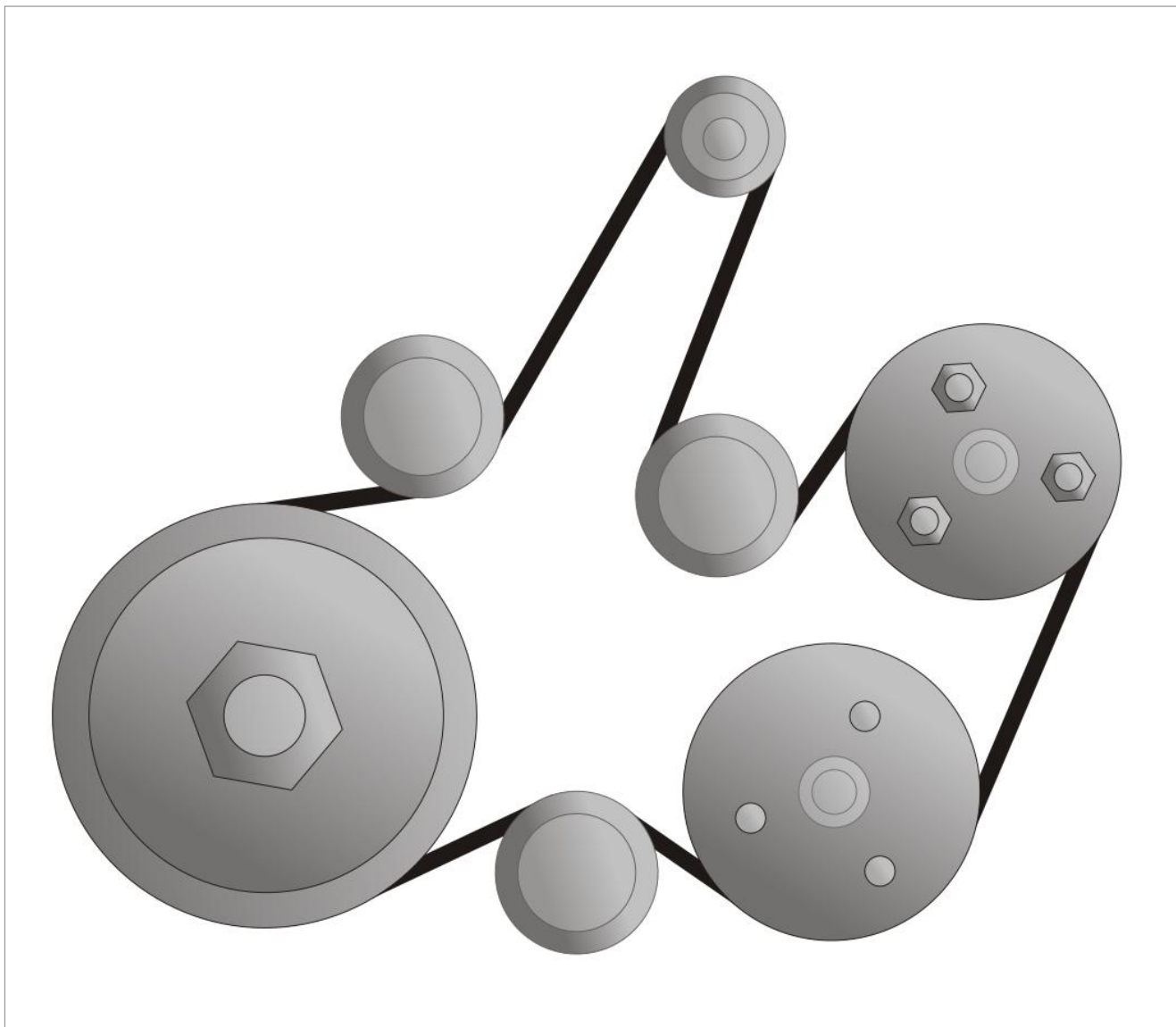
Remove the ancillary drive belt

Drive belt layout (Standard suspension)



Drive belt layout (With Dynamic Response Suspension)

Repair Manuals

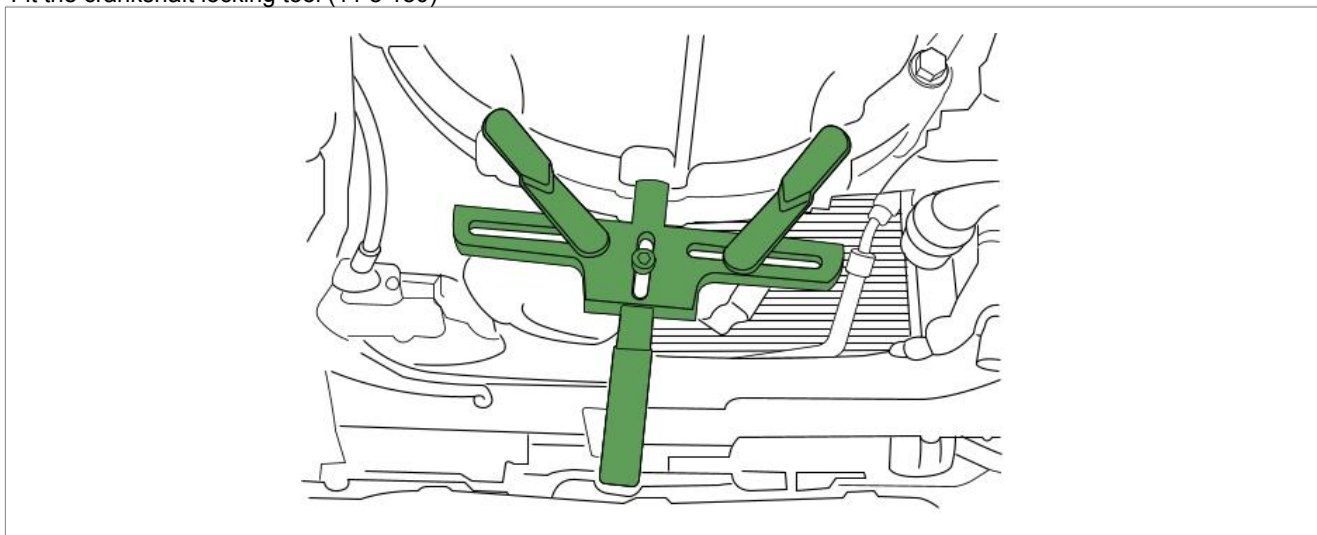


Remove the ancillary drive belt tensioner

Remove the underbody cover

Vehicles with manual gearbox:

Fit the crankshaft locking tool (11 8 180)



Repair Manuals

Vehicles with automatic transmission:

Fit the crankshaft locking tool (11 8 660)

All vehicles:

Remove the vibration damper

Note: Do not loosen the crankshaft gearwheel centre bolt

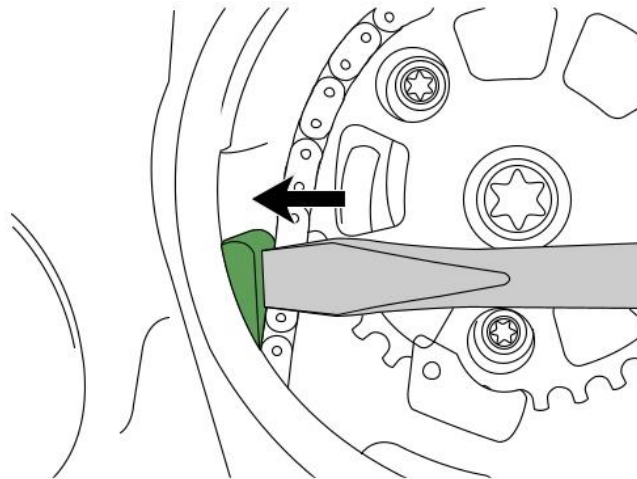
Vacuum pump:

Remove the sealing cap

Oil pump:

Remove the sealing cap

Loosen the chain tensioner piston



Fit the tensioner locking pin

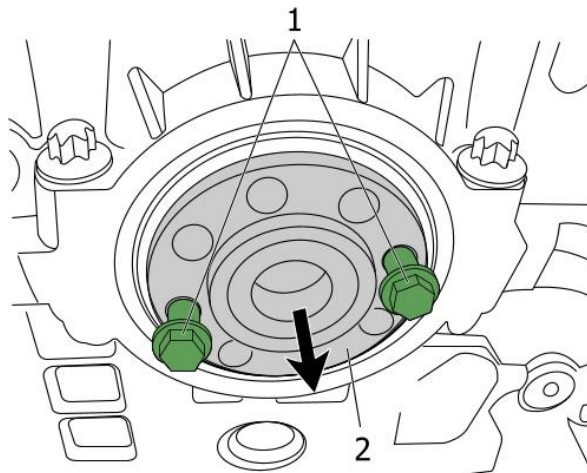
Lock the tensioner

Remove the crankshaft gearwheel centre bolt

Crankshaft pulley outer bolts:

Refit the bolts shown (1)

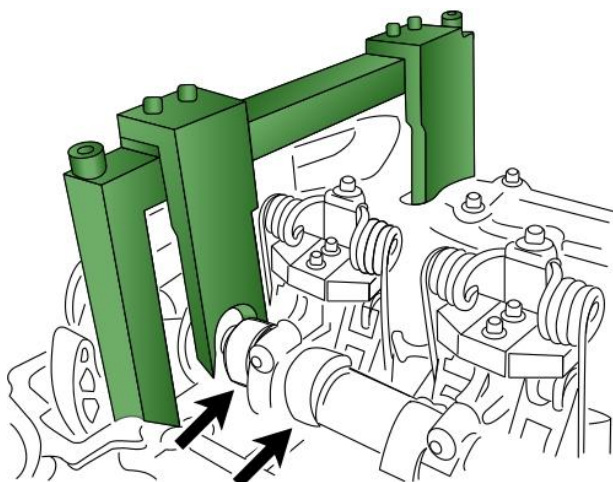
Remove the crankshaft hub (2)



Remove the timing chain guide bolts from the cylinder head

Fit the camshaft locking tools (11 4 281, 11 4 282, 11 4 283)

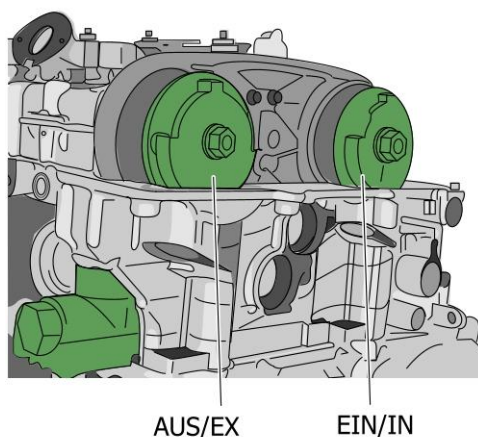
Repair Manuals



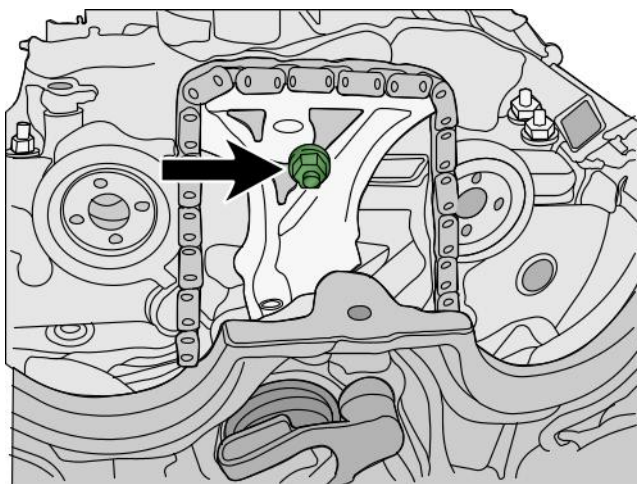
Remove the upper chain tensioner

Remove the inlet camshaft timing adjuster (EIN/EI)

Remove the exhaust camshaft timing adjuster (AUS/EX)



Remove the timing chain guide bolts from the cylinder head (x 2)



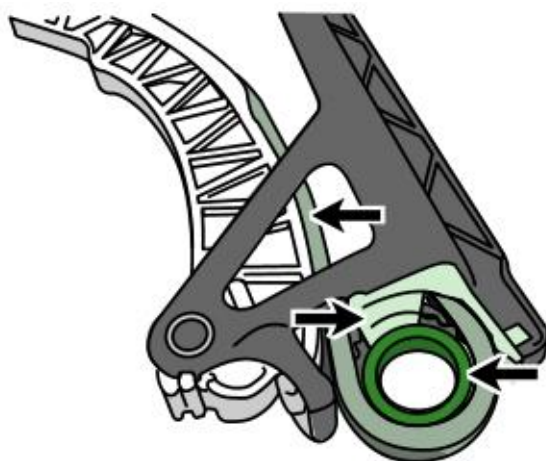
Remove the bearing pins for the chain guide rail and the tensioner rail

Remove the chain guide with the timing chain, guide rail and tensioner rail

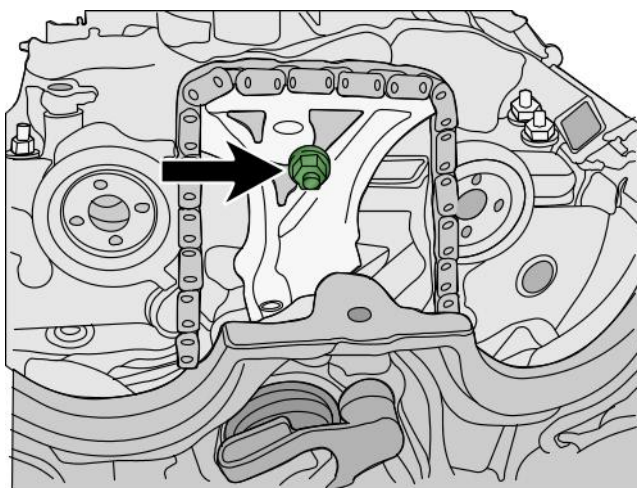
Installation

Refit the chain guide with the timing chain, guide rail and tensioner rail

Repair Manuals



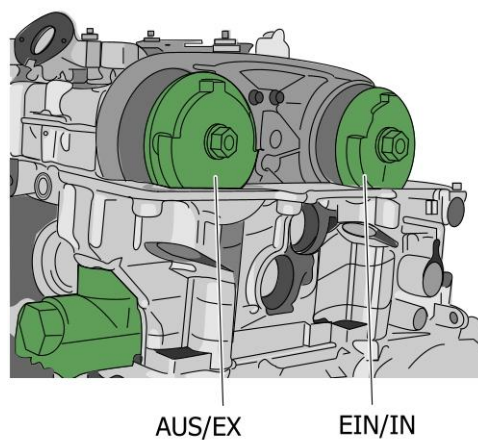
Refit the timing chain guide bolts to the cylinder head (x 2)



Refit the crankshaft hub

Refit the exhaust camshaft adjuster (AUS/EX)

Refit the inlet camshaft timing adjuster (EIN/IN)

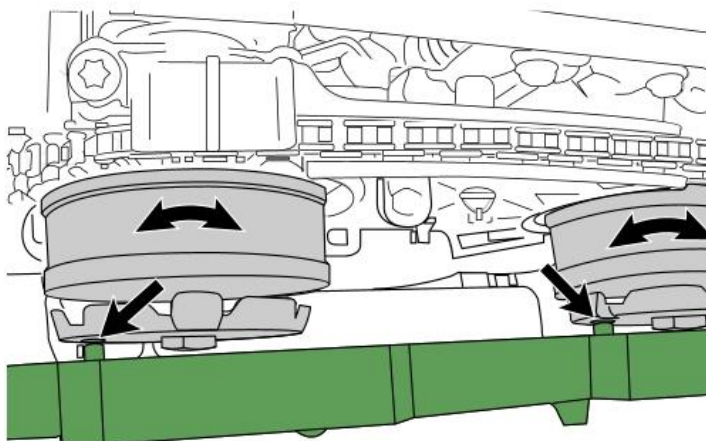


Tighten the camshaft gearwheel bolts lightly until there is no play

Fit the camshaft alignment tool (11 4 290)

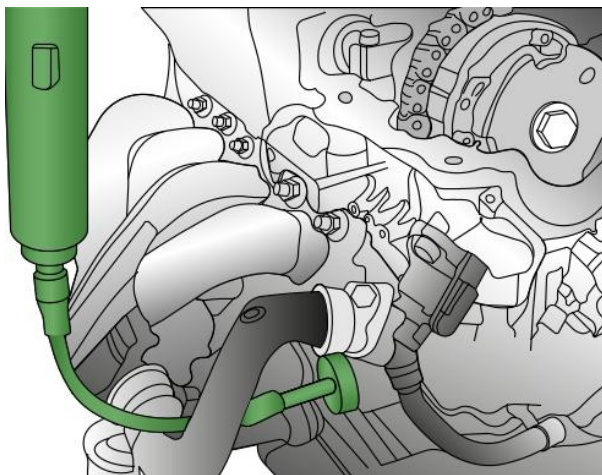
Rotate the camshaft sensor wheels until the retaining pins on the special tool fit correctly

Repair Manuals



Pre-tension the timing chain (0.6 Nm)

Use the special tools (00 9 250, 11 9 340)



Tighten the camshaft gearwheel bolts

Installation is in the reverse order of removal

Torque settings

Timing chain guide

Timing chain guide (Lower) 20 (Nm)

Timing chain guide (Upper) 14 (Nm)

Timing chain tensioner

Timing chain tensioner (Upper) 55 (Nm)

Renew the seal (Threads lightly oiled)

Crankshaft pulley

Crankshaft pulley (Outer bolts) 35 (Nm)

Crankshaft gearwheel

Crankshaft gearwheel (Centre bolt)

Renew the bolt (Threads lightly oiled)

Stage 1 100 (Nm)

Stage 2 270 (°)

Valve cover

Valve cover 8.5 (Nm)

Spark plugs

Spark plugs (M12) 20 - 26 (Nm)

Spark plugs (M14) 27 - 33 (Nm)

Repair Manuals

Camshaft gearwheel

Camshaft gearwheel

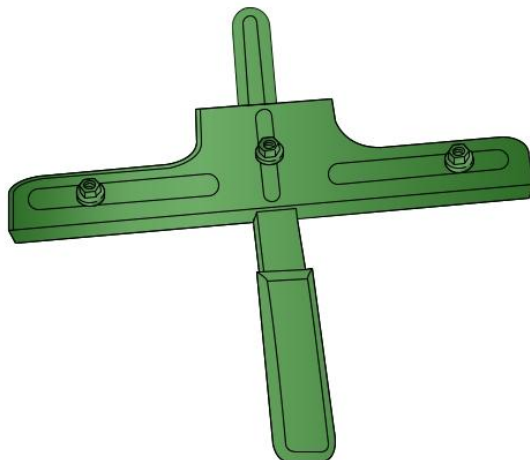
Renew the bolts

Stage 1 20 (Nm)

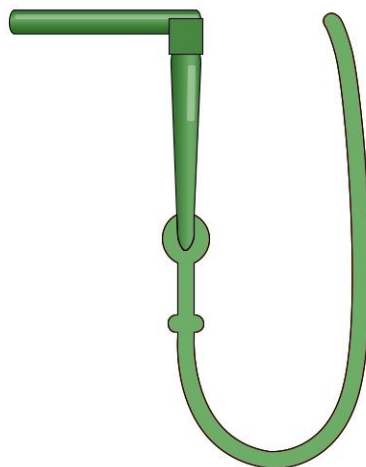
Stage 2 180 (°)

Special tools

Crankshaft locking tool: (11 8 180)

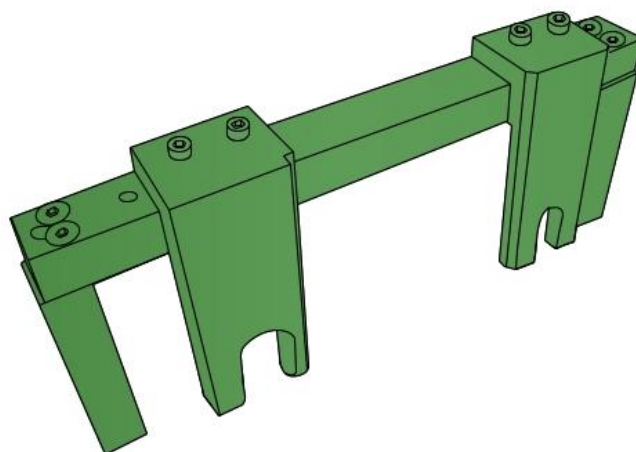


Crankshaft locking tool: (11 0 300)

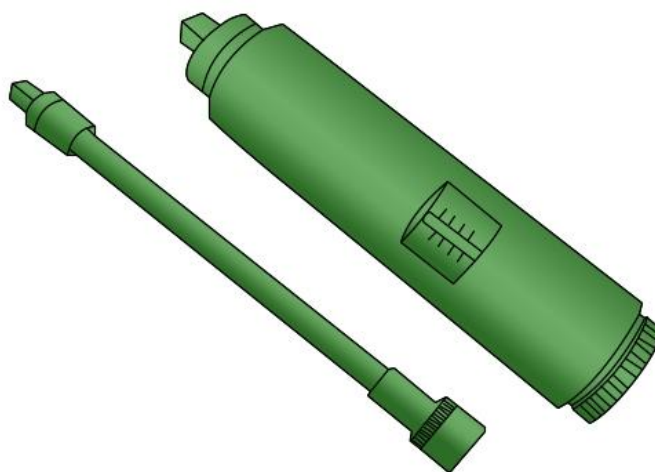


Camshaft locking tool: (11 4 281, 11 4 282, 11 4 283)

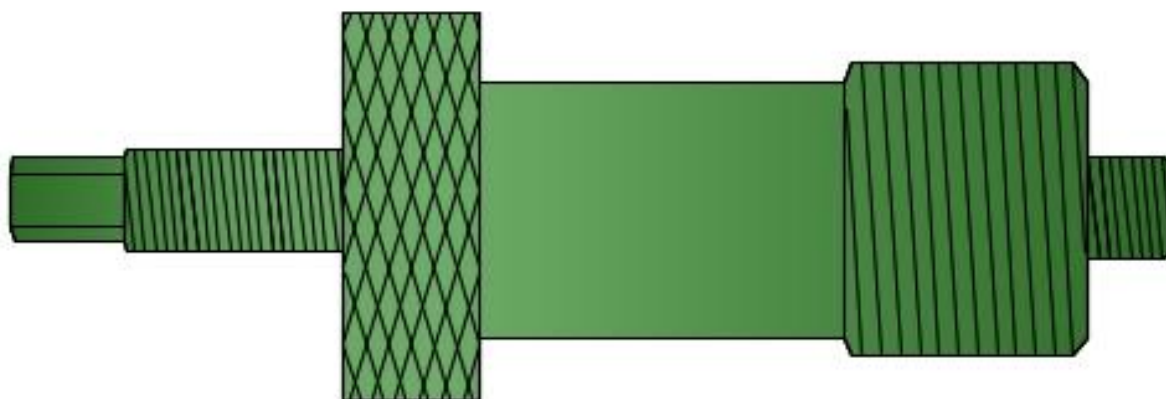
Repair Manuals



Adjustment tool: (00 9 250)

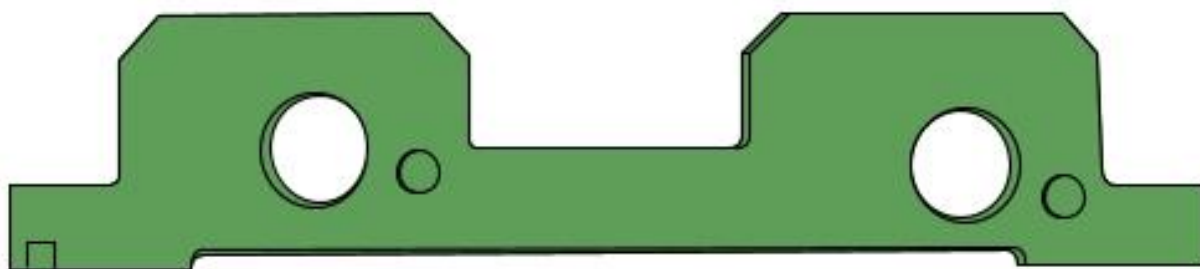


Adaptor: (11 9 340)



Camshaft alignment tool: (11 4 290)

Repair Manuals



Flywheel locking tool: (11 8 660)

